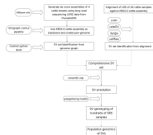


## Masculine Leadership

## 2 Material and Method



- Short reads sequences (SRS) : 18 breeds, 234 samples
- Long reads sequences (LRS) : 15 breeds, 45 samples
- Publicly available SRS : 25 breeds, ~500 samples

| Breed             | length (in<br>gbp) | contigs | N50 (in<br>~Mbp) | L50 |
|-------------------|--------------------|---------|------------------|-----|
| Brachieros        | 3.29               | 446     | 91.5             | 14  |
| Morocha           | 3.19               | 1064    | 72.7             | 16  |
| Bulgarian<br>Gray | 3.21               | 1137    | 63.6             | 17  |
| Podolian          | 3.19               | 1400    | 63.2             | 18  |

Horizontal stacked bar chart showing the number of reads for four RNA-seq datasets (lygR01, rsmR01, grrR01, pcdR01) across four categories: Complete (S) and single-pass (S), Complete (S) and duplicated (S), Fragmented (S), and Missing (S). The x-axis represents the number of reads from 0 to 100. The legend indicates: Complete (S) and single-pass (S) in blue, Complete (S) and duplicated (S) in dark blue, Fragmented (S) in yellow, and Missing (S) in red.

| Dataset        | Complete (S) and single-pass (S) | Complete (S) and duplicated (S) | Fragmented (S) | Missing (S) |
|----------------|----------------------------------|---------------------------------|----------------|-------------|
| lygR01_rnamRNA | 85.928                           | 0.140                           | 1.24           | 0.274       |
| rsmR01_rnamRNA | 89.40                            | 0.180                           | 0.21           | 0.256       |
| grrR01_rnamRNA | 89.64                            | 0.175                           | 0.23           | 0.285       |
| pcdR01_rnamRNA | 89.40                            | 0.178                           | 0.22           | 0.288       |

#### 4 Next steps:

- Construct a cattle pangenome using the *de novo* assemblies and the ARS-UCD2.0 reference genome.
- Identify breed-specific structural variants (SVs) using graph-based approaches.
- Compare pangenome-based and alignment-based SV discovery methods.
- Perform SV annotation and population genomic analyses for functional characterization.